

Reconstructing Time-Resolved Wake Dynamics from Sparse PIV-like Measurements: A Data- Driven Approach

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Setup and Motivation

- **Challenge:** Reconstructing non-time-resolved PIV data due to prohibitive high-speed equipment costs.
- **New Proposed Approach:** Reservoir Computing (RC) to parameterize hidden memory effects without fine-scale simulations.
- **Validation:** 2D FEM simulations (FEniCSx) of flow around a fluidic pinball.

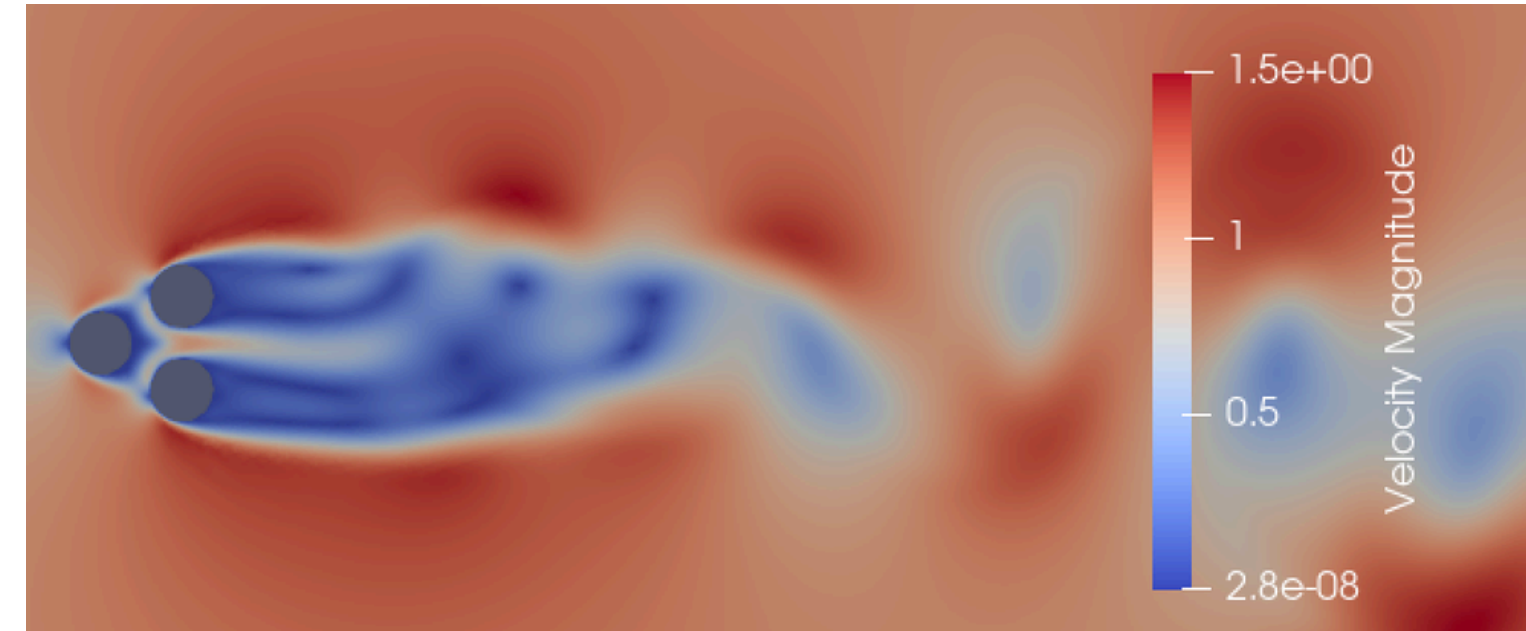


Fig 1 - Example pinball data snapshot

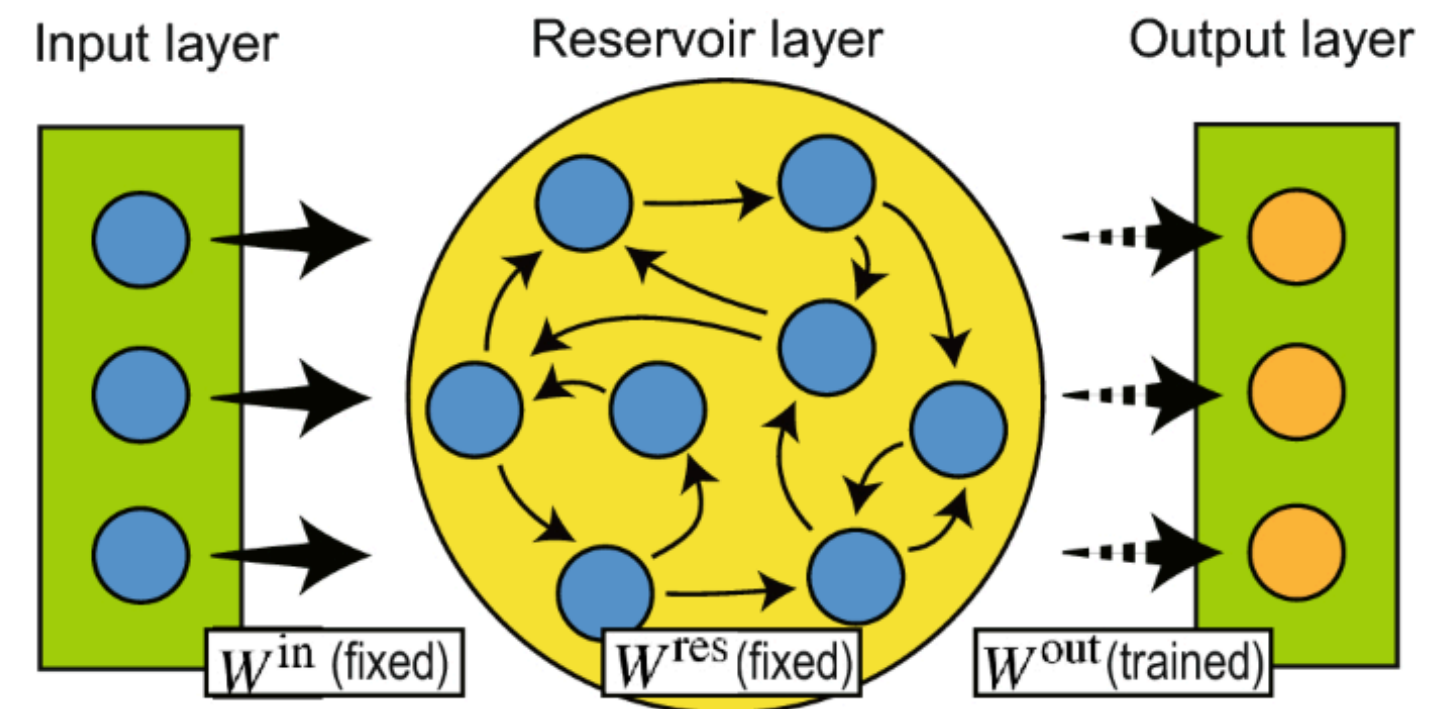


Fig 2 - Reservoir illustration:

<https://www.nature.com/articles/s41598-020-78725-0>

Reservoir Computing Overview and Results

- **Enhanced Reconstruction:** RC captures non-Markovian dynamics and significantly reduces closure error.
- **Superior Performance:** Outperforms standard Markovian baselines (LSE, KF/KS).
- **Key Takeaway:** Flow history is critical; RC effectively models unresolved scales.

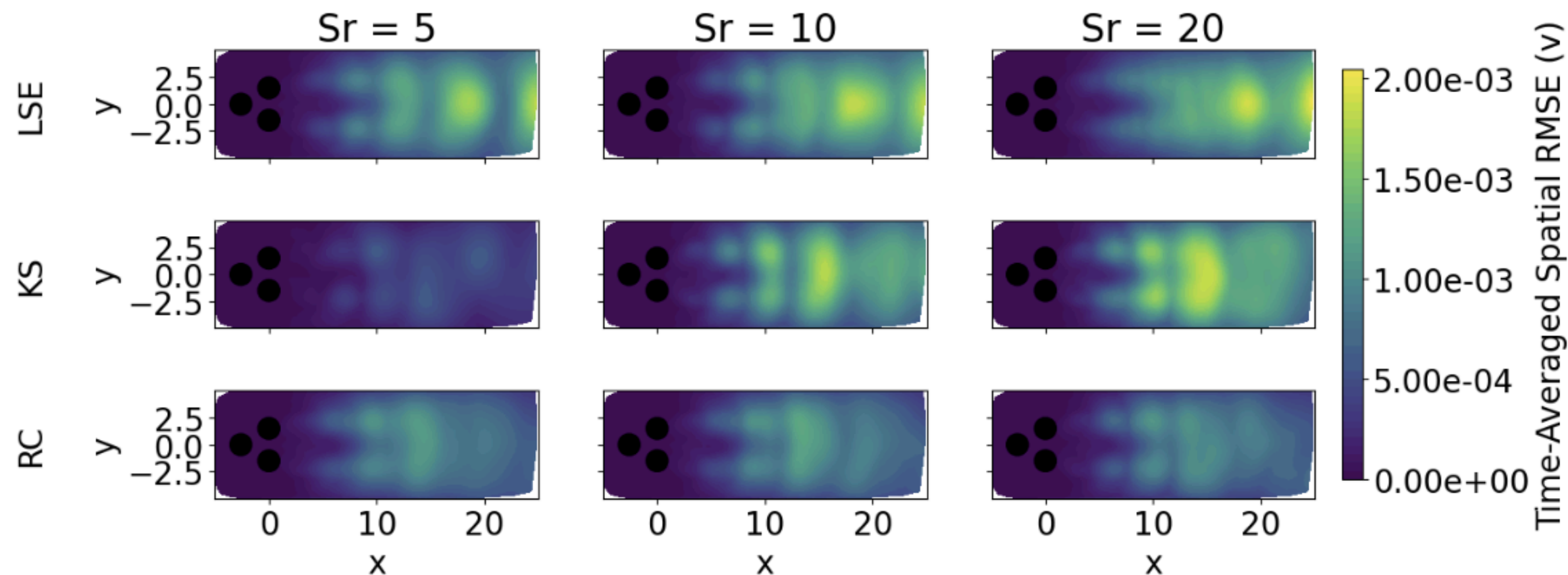


Fig 3 - RMSE of standard methods (Linear Stochastic Estimator and Kalman filtering and smoothing) compared to Reservoir computing for different under-sampling cases (higher Sr is harder to reconstruct)

Thank you !
Visit me at my poster :)